

PrefLens: Position Artefacts and Convergent Validity in LLM Preference Elicitation

A multi-method study, a controlled within-family scale follow-up, and a framing test

PrefLens — Digital Minds Research Sprint, Track 4 (Preference Elicitation Methods) Code and raw data: <https://github.com/hajraRehman/PrefLens> Run date: 2026-08-16. All reported results are reproducible from the committed raw-data directories (data/raw/{pilot,main,manipulation_check,followup_gpt_oss,followup_gpt_oss_provider_pinned,followup_gpt_oss_neutral_framing}/) and the configs in configs/.

Abstract

LLM preference measurement has no direct ground truth, so convergent validity across elicitation procedures is one useful source of evidence about measurement consistency. Study 1 applied four distinct procedures to 12 welfare-neutral task pairs across three model families (4,013 trial records). Against a matched permutation null, only Gemini-3.1-flash-lite converged above chance ($p = 0.0019$); Qwen-2.5-7B's pairwise scores reproduced the display-order draw exactly on all 12 items. In an exploratory self-report/pairwise reanalysis, equal-weighting the two display orders raised Llama's observed correlation from -0.007 to $+0.524$, suggesting order imbalance can obscure cross-method agreement. Study 2b (exact counterbalancing, provider-pinned) found the larger GPT-OSS model *more* position-dominated ($\Delta = -0.333$), rejecting our pre-specified hypothesis. Extreme position sensitivity in GPT-OSS 120B persisted after removing the prompt's explicit indifference cue; the smaller model showed a suggestive framing effect. Preference studies should counterbalance option order and report position-conditioned rates.

(142 words)

1. Introduction

Work on apparent LLM preferences typically operationalises "preference" one way — most often repeated pairwise forced choice — and reports the resulting numbers. The field has no ground truth: there is no independently verified fact about what a model prefers against which an elicited score could be scored correct.

In the absence of direct ground truth, **convergent validity** provides one useful source of evidence about measurement consistency (others include test-retest reliability, controlled interventions, and synthetic tasks where ground truth is constructed). If several distinct operationalisations of the same construct land in the same place, the estimate is at least method-robust. If they do not, conclusions drawn from any single procedure are conclusions about the procedure as much as about the model.

These procedures are **distinct but not mechanistically independent**: all query the same model through the same text channel, and share its post-training and response heuristics. Agreement between them can therefore reflect a shared bias rather than a common signal.

This work treats preference elicitation as a measurement-validity problem, in three studies: a multi-method convergence study, a within-family scale comparison (with a provider-controlled replication, Study 2b), and a test of whether the elicitation prompt's own framing drives the artefact.

RQ1. Do multiple elicitation methods converge on the same apparent preferences? (Study 1: four methods, three model families.)

RQ2. How strongly can display position distort preference measurement? (Diagnostic analysis within Study 1.)

RQ3. Does positional susceptibility differ with model scale *within* one family? (Studies 2/2b: GPT-OSS 20B vs 120B, exact counterbalancing; **Study 2b** provides the provider-pinned comparison.)

RQ4. Does the elicitation prompt's own framing change positional susceptibility? (Study 3: Study 2b with the indifference cue deleted.)

Study 1 raised RQ3 but cannot answer it: its three models differ in family, scale, data, post-training and serving stack simultaneously. **Study 2b** substantially reduces these confounds by holding model family, upstream inference provider, prompts, sampling, task set and protocol fixed while comparing two model scales. (The original Study 2 did not pin the upstream provider; see §6.4.)

Contributions.

- 1. A reusable, fully logged elicitation harness implementing four methods behind one provider-agnostic interface, with per-trial order randomisation, strict parse accounting, checkpointing, and a full test suite.
- 2. A cross-method convergence analysis on three model families, tested against a **matched permutation null** that preserves each procedure's observed score distribution and destroys only cross-method item alignment.
- 3. **A position/content decomposition that distinguishes cross-method disagreement from cases where an elicitation measure carries no detectable order-invariant content signal** — and the finding that one standard pairwise elicitation was entirely position artefact.
- 4. Two pre-specified checks that changed the design mid-study: a manipulation check that invalidated our original trade-off scoring, and a degeneracy check that invalidated one of our own computed results.

We make no claim about consciousness, sentience, welfare, or genuine internal preference. See the Limitations appendix.

2. Related work

Our pairwise method is a stripped-down independent re-implementation of the forced-choice elicitation used in **utility engineering** work (Mazeika et al., 2025), which samples pairwise preferences repeatedly and fits utility models, reporting that preference structure becomes more internally coherent with scale. We build on it as prior methodology and baseline; no external code is vendored.

Our contribution is not the discovery of pairwise LLM preferences. It is the convergent-validity question: whether *several* operationalisations of the same apparent preference agree, and what it means when they do not.

Position and selection bias in LLM multiple-choice responding is well documented (Pezeshkpour & Hruschka, 2023; Zheng et al., 2024), including token-level bias toward specific option IDs. Our results connect that literature directly to preference elicitation: an unreported position bias does not merely add noise, it can constitute the *entire* measured signal. Notably, Llama-3.1-8B-Instruct has independently been reported to show strong primacy bias toward option A (arXiv:2605.01846), consistent with what we observe.

3. Study 1 — Multi-method preference elicitation: methodology

3.1 Models

key	model ID (pinned)	provider	family
llama31-8b	meta-llama/llama-3.1-8b-instruct	OpenRouter	Llama
qwen25-7b	qwen/qwen-2.5-7b-instruct	OpenRouter	Qwen
gemini-31-flash-lite	gemini-3.1-flash-lite	Google	Gemini

IDs are pinned to explicit versions; OpenRouter calls disable provider fallbacks; the served model is recorded on every record and was verified to match the request on 100% of calls. Alias endpoints were rejected because they can resolve to different models between calls (D-02, D-22).

Sampling: temperature 1.0, top_p 1.0, max_tokens 300, identical across models. Temperature is non-zero deliberately: Methods B–D estimate a *choice probability*, which is degenerate at temperature 0 (D-05).

3.2 Preference set

12 analysis items across 5 categories (cognitive style, task structure, interaction style, information processing, creation vs evaluation), plus 2 sanity controls pairing a coherent option against a degenerate one. Constraints are enforced by tests, not by eye: option strings within 1.20× on characters and 3 words, no evaluative/helpfulness/safety/distress vocabulary, documented per-item confounds. Full set in `configs/preferences.yaml`.

3.3 Methods, and normalisation to a common scale

Every method reduces to a signed score in `[-1, +1]` where +1 is maximal evidence for semantic option A and 0 is indifference.

Method	Estimand	Normalisation
A Direct self-report	stated choice + stated strength	mean of sign × strength
B Repeated pairwise	choice frequency	$2 \cdot P(A) - 1$
C Cost trade-off	choice across a symmetric cost ladder	$2 \cdot \text{mean_levels} \cdot P(A) - 1$
D Sequential selection	share of episode stages spent on A	$2 \cdot \text{occupancy} - 1$

A test asserts all four return exactly +1 for the same underlying behaviour.

Method C was rescored mid-study. It originally estimated an indifference point (the interpolated $P(A)=0.5$ crossing of a signed cost ladder). A manipulation check — *both options identical*, surcharge on one, isolating the cost clause from item preference — showed that assumption fails:

model	avoid rate	p (vs. no effect)
llama31-8b	0.700	0.021
qwen25-7b	0.567	0.292

The surcharge deters on Llama but is **flat in magnitude** (avoid rate ≈ constant from cost 1 to 16), and on Qwen has no detectable effect at all. Against a saturating curve the 0.5 crossing is located by sampling noise. Method C is therefore scored as the mean over all nine ladder rungs — robust to non-monotonicity — with the indifference point retained only as a diagnostic (D-20). Consequently **Method C on Qwen is a degenerate case**: the surcharge is inert, so the procedure reduces to a noisier Method B, and "willingness to trade" is not claimed for that model.

3.4 Randomisation and controls

Display order is randomised per trial from a reproducible seed; the semantic ↔ displayed mapping is undone in exactly one function, so the parser cannot introduce a position artefact. Realised balance was 0.46–0.57 $P(\text{order}=ab)$ across all model × method cells. Framings (neutral / preference / action) alter only the question sentence, for Methods A and B (D-07). Sanity controls are reported separately and excluded from all convergence metrics.

3.5 Metrics

Direction agreement (with a $|s| \leq 0.05$ dead zone), Spearman ρ with percentile bootstrap CIs over items, Pearson r as supplementary, mean absolute disagreement, sign-flip rate, and a bounded composite (CMCS) reported *alongside* — never instead of — the standard metrics. Because these statistics have a non-trivial value even with no cross-method alignment, every observation is tested against a **matched permutation null** (10,000 permutations). Each method column retains its observed values — its distribution, ties, saturation and dead-zone behaviour — while item labels are permuted independently within each column. The only structure destroyed is the alignment of items across methods, which is exactly the hypothesis under test. The observed value and every permutation draw are produced by the same estimator function. (An earlier parametric baseline was withdrawn during final audit; see D-32 and §4.1.)

Position/content decomposition. For each (model, method, item):

```
position = P(A | A shown first) - P(A | A shown second)
content  = mean of the two - 0.5
```

content is the order-free preference estimate; averaging the two display orders cancels a symmetric position effect. A measure is flagged **degenerate** when no item shows any content.

3.6 Execution and data quality

Quantities are defined precisely, because an earlier draft quoted a "4,232 calls" total that reconciles to no defined quantity and has been withdrawn (D-36). All figures below are computed from the committed artefacts by `src/claims.py`.

quantity	Study 1
committed trial records (rows in the raw JSONL)	4,013
of which pilot / main / manipulation check	116 / 3,717 / 180
successful model responses (<code>call_ok</code>)	3,928
failed calls	85
API attempts including retries	4,533

Of the 3,437 choice-turn records (excluding free-text `perform_task` turns):

- **85 call failures (2.5%)** — all Gemini HTTP 429 after the free-tier daily cap of 500 requests was exhausted; zero failures on Llama and Qwen.
- **1 parse failure (0.03%)** — Llama returning `{"choice": "Both are equally valid options"}` on a control item.
- **99.7% strict JSON**; the remainder recovered from prose and logged as such.

The Gemini quota truncated the run **mid-queue**, so completed cells are complete: Methods A/B on 11 items, Method C on 10, **Method D not started** (D-24). Because agreement statistics depend mechanically on the number of methods and items, **all cross-model comparison uses a matched subset: 3 methods × 10 items common to every model** (items whose every cell is fully observed in every model; a cell below 80% of its planned observations is set to missing, so an interrupted run cannot inflate agreement).

4. Study 1 results

4.1 Cross-method convergence differs sharply by model

Matched subset (Methods A, B, C; 10 items):

model	direction agreement	mean ρ	sign-flip rate	MAD	all-3 agree	CMCS
gemini-31-flan-0.911ite	0.911	+0.868	0.089	0.303	9/10	0.803
llama31-8b	0.690	+0.308	0.310	0.330	6/10	0.785
qwen25-7b	0.565	+0.021	0.435	0.366	5/10	0.777

The null. Convergence statistics have a non-trivial value even with no cross-method alignment, so each observation is compared against a **matched permutation null** (10,000 permutations). Each method column keeps its observed values — distribution, ties, saturation, dead-zone behaviour and all — while item labels are permuted independently within each column. This destroys exactly one thing, the alignment of items across methods, and nothing else. The null and the observation are computed by the *same* estimator function, averaged over all k-choose-2 method pairs.

An earlier parametric baseline (simulating every method as fair coin flips) was found during final audit to be mis-specified in three ways and has been withdrawn (D-32). Its numbers do not appear in this report.

Matched subset (3 methods × 10 items):

model	statistic	observed	null mean	null 95th	empirical p
gemini-31-flash-lite	direction agreement	0.911	0.568	0.732	0.0019
gemini-31-flash-lite	mean p	0.868	−0.002	0.346	0.0001
llama31-8b	direction agreement	0.690	0.556	0.746	0.1231
llama31-8b	mean p	0.308	0.002	0.342	0.0693
qwen25-7b	direction agreement	0.565	0.501	0.694	0.2736
qwen25-7b	mean p	0.021	0.001	0.345	0.4325

On the matched basis, **only Gemini exceeds its permutation null**, and it does so decisively on both statistics. Llama does not ($p = 0.123, 0.069$); Qwen does not.

A qualification we must state. On each model's *fuller* basis — Llama and Qwen have four methods and 12 items, Gemini only three and 11 — Llama's convergence *does* exceed its own null nominally (direction agreement 0.712, $p = 0.0140$; mean p 0.255, $p = 0.0328$). With six model $\times$ statistic tests, a Bonferroni threshold is 0.0083, which Llama's values do not meet and Gemini's (0.0011, 0.0001) do. We therefore report Llama as **suggestive but not established**: significant on one basis at nominal α , not on the matched basis, and not after correction for multiplicity. The earlier flat null claim for Llama was an artefact of the withdrawn baseline (D-32) and is not repeated.

Note CMCS ≈ 0.78 – 0.80 for *all three models*, and its permutation null sits in the same range — it is uninformative here, exactly the failure mode anticipated when defining it: it rewards agreement near indifference. This is why five standard metrics were reported alongside it.

4.2 The explanation: convergence tracks position-independent signal

model	method	mean content	median	max	mean position	degenerate
gemini	pairwise	0.325	0.400	0.500	+0.305	no
gemini	self-report	0.293	0.375	0.500	+0.280	no
llama	pairwise	0.185	0.119	0.500	+0.318	no
llama	self-report	0.079	0.036	0.417	+0.841	no
llama	sequential	0.197	0.125	0.500	+0.424	no
qwen	pairwise	0.000	0.000	0.000	+1.000	YES
qwen	self-report	0.156	0.000	0.500	+0.688	no
qwen	sequential	0.015	0.000	0.167	+0.970	no

The distribution is reported continuously. An earlier version of this table counted how many items exceeded $|\text{content}| > 0.15$; that cut had no derivation and was not pre-specified, so the categorical column was removed (D-35). *degenerate* is a numerical condition — every item's content effect equal to zero within tolerance — not a hypothesis test (D-34).

Pooled position effects were large in every model: $\Delta = +0.292$ (Gemini), **+0.509** (Llama), **+0.762** (Qwen), all $p < 10^{-10}$.

Qwen's pairwise measure is pure position artefact. On all 12 items, $P(\text{semantic } A)$ equalled the fraction of trials in which A happened to be displayed first — $0.3 \rightarrow 0.3$, $0.7 \rightarrow 0.7$, $0.4 \rightarrow 0.4$, and so on. The score encodes the random display-order draw and nothing else. Qwen still discriminated the sanity controls (mean $+0.67$ to $+0.93$), so it is not

incoherent. Under this protocol its choices among welfare-neutral items showed **no detectable order-invariant differential signal** and were fully accounted for by display position. That is a statement about what this measurement recovered, not about whether the model has preferences.

Across the three models, convergence rises monotonically with mean |content| (Figure 5). With **only three models this is a suggestive association, not an established relationship** — three points cannot support a quantitative claim.

4.3 Exploratory: does removing position change the convergence conclusion?

If some raw scores carry large position effects, part of what looks like *methodological* disagreement may be nuisance variance. We can test this with no new data by averaging the two display-order conditions, which cancels a symmetric position effect:

s_adjusted = (E[s | semantic A shown first] + E[s | semantic A shown second]) / 2

Scope. This is exact only for self-report and pairwise. The trade-off score averages $P(A)$ *within* each of nine cost rungs and 22.7% of (item × rung) cells observed only one order; the sequential score is occupancy across three independently randomised stages, so an episode has no single display order. Rather than impute either, both are excluded (D-39). Adjusted convergence below is therefore a **single method pair**, not an average over pairs, and is not comparable in magnitude to the 3-method figures in §4.1. Raw and adjusted are computed on the same two-method basis.

Estimator parity. raw is taken directly from analysis.build_scores — the same score the rest of the paper uses — not recomputed here, and adjusted reuses the identical self-report imputation constant. A check asserts the two agree to 1.1e-16 per cell. An earlier version recomputed the imputation median from the neutral-only subset and was therefore comparing two slightly different estimators (D-41). Note the basis here is each model's full item set (11–12 items), so the raw values differ from the 10-item matched subset in §4.4 for that reason alone.

model	raw ρ	p (perm)	adjusted ρ	p (perm)	raw dir. agr.	adjusted dir. agr.
Gemini	0.885	0.0006	0.924	0.0003	0.875	1.000
Llama	−0.007	0.5101	+0.524	0.0420	0.571	0.833
Qwen	−0.190	0.7263	undefined	—	0.600	undefined

For Llama's self-report/pairwise pair, **removing realised order imbalance increased the observed correlation substantially** ($\rho = -0.007 \rightarrow +0.524$; direction agreement $0.571 \rightarrow 0.833$), which is consistent with display position contributing to their apparent disagreement. We have not tested whether the *change* itself is statistically significant — the quoted p-values are permutation tests of each correlation against its own null, not of the difference between them.

Gemini was already converged and adjustment changes little (+0.040).

Qwen is undefined, and that is the result. Its adjusted pairwise score is exactly 0.000 on all twelve items — a constant vector, for which a rank correlation does not exist. Once display order is removed, nothing remains of that measure to correlate with anything.

This analysis is **exploratory**, generated by the diagnostics rather than pre-specified, and rests on a single method pair over 11–12 items. It does not license a claim about §4.1's three-method figures, which include two procedures this adjustment cannot be applied to. What it does show is narrower and still useful: for one method pair in one model, order imbalance was large enough to mask a moderate correlation, so a convergence study that leaves position in the estimate can **understate** agreement as well as manufacture it.

4.4 Which method pairs agree

pair	Gemini ρ [95% CI]	Llama ρ [95% CI]	Qwen ρ [95% CI]
self-report / pairwise	+0.859 [0.451, 0.990]	+0.037 [−0.769, 0.769]	−0.130 [−0.793, 0.575]

self-report / trade-off	+0.924 [0.555, 1.000]	+0.201 [-0.596, 0.811]	+0.451 [-0.349, 0.997]
pairwise / trade-off	+0.820 [0.389, 0.977]	+0.687 [0.106, 0.962]	-0.258 [-0.818, 0.456]

On Llama, the only pair whose CI excludes zero is **pairwise/trade-off** ($p = +0.687$) — and that agreement is partly artefactual, since the weak cost manipulation makes Method C partially reduce to Method B. Discounting it, Llama shows no method pair that convergently validates another. Direct self-report is the weakest partner for the behavioural methods on both open-weight models.

4.5 Secondary hypothesis: contradicted

We predicted stronger pairwise preferences would be *more* method-stable (negative ρ between $|\text{pairwise strength}|$ and cross-method dispersion). Observed:

model	ρ	95% CI	p	valid
gemini	+0.621	[0.018, 0.932]	0.041	yes
llama	+0.448	[-0.268, 0.813]	0.144	yes
~~qwen~~	~~+0.592~~	—	~~0.043~~	NO — withdrawn

Both valid estimates are **positive**, i.e. the opposite of our hypothesis: stronger pairwise margins went with *more* cross-method disagreement. Only Gemini's is nominally significant, on 11 items, and we treat it as suggestive only.

The Qwen result is withdrawn. H2 uses $|\text{pairwise strength}|$ as its x-axis; for Qwen that quantity is the random display-order draw, so the apparently significant $p = 0.043$ is an artefact of correlating noise with dispersion. It was computed, detected by the degeneracy check, and is reported here as withdrawn rather than deleted.

4.6 Framing sensitivity

Changing only the question sentence flipped preference direction on a substantial minority of items:

model	method	flip rate range	mean $ \Delta\text{score} $ range
llama	self-report	0.167 – 0.429	0.195 – 0.258
llama	pairwise	0.125 – 0.333	0.317 – 0.333
qwen	self-report	0.222 – 0.556	0.191 – 0.298
qwen	pairwise	0.222 – 0.500	0.300 – 0.450

"Which would you *select*" versus "which would you *prefer*" — semantically near identical, alternatives byte-identical — changed the inferred direction on up to **56%** of items. Framing robustness was not tested for Methods C and D (D-07).

4.7 Cross-model replication

Per-method Spearman ρ between Gemini and Llama item scores ($n = 11$): self-report +0.178 [-0.532, 0.780] ($n=11$); pairwise +0.463 [-0.260, 0.920] ($n=11$); trade-off +0.616 [-0.241, 0.978] ($n=10$). **All CIs include zero.** We find no evidence that the models rank these items alike, though 10–11 items cannot rule it out either.

5. Study 2 — Initial within-family position-bias follow-up: methodology

5.1 Motivation

Study 1's diagnostic found that cross-method convergence tracked how position-independent each model's choices were, and that Qwen's pairwise measure was pure position artefact. An obvious explanation is capability: perhaps small models fall back on "pick the first one" because they cannot compare the options.

Study 1 cannot test that. Qwen, Llama and Gemini differ in family, scale, training data, post-training, architecture **and serving provider** at once.

Scale nevertheless remains confounded with training compute, data mixture and post-training, which differ between sizes even within a family.

5.2 Design

	mopenai/gpt-oss-20b, openai/gpt-oss-120b
OpenRouter, both models, standard paid endpoints	
upstream inferred, provider recorded	
— repaired in Study 2b (\$6.4)	
items same 12	
balanced items as Study 1, plus the 2 sanity controls	
design items x 2 positions x 10 repetitions x 2 models = 480	
principal trials (+80 control)	

temperature	1.0,
top_p	1.0,
max_tokens	800,
identical	across
arms	
prompts	forced
choice;	no
justification,	no
introspection	requested

A :free endpoint exists for the 20B model but not the 120B. We originally used the standard paid endpoint for both, intending to reduce serving-stack differences. **Final audit showed this was insufficient**: routing through one gateway does not pin the upstream inference provider, which remained uncontrolled and unrecorded. Study 2b repairs this (\$6.4). Total cost: under one US cent.

5.3 Exact counterbalancing

Study 1 randomised display order probabilistically, which is unbiased in expectation but leaves realised splits unequal (0.46–0.57 in Study 1). When the position effect *is* the estimand, that imbalance is avoidable noise placed directly on the quantity of interest.

Study 2 uses **exact counterbalancing**: for every (item, model), exactly 10 trials place semantic option X first and exactly 10 place it second. Balance is structural rather than probabilistic, and is verified before any statistic is computed (D-28).

5.4 Estimands

For semantic option X of each item:

```
p_first = P(select X | X displayed first)
p_second = P(select X | X displayed second)

position_effect = p_first - p_second      in [-1, +1]
content_signal = p_first + p_second - 1   in [-1, +1]
```

These are the orthogonal rotation of (p_first, p_second): the first cancels content, the second cancels a symmetric position effect. content_signal is an **order-invariant content-associated signal** — it means choices covaried with which task was described. It is not a genuine, true or internal preference, and is never described as one (D-29).

Uncertainty is a percentile bootstrap over **items** (10,000 resamples), since the generalisation claim is about preference items, not about individual responses. The two arms are the same items measured twice, so the difference bootstrap resamples items **jointly (paired)**.

5.5 Pre-specified hypotheses

Recorded in DECISIONS.md (D-27) **before any Study 2 data was collected**:

- **H1** — the larger model will show a *smaller* mean |position effect|.
- **H2** — the larger model will show a *larger* mean |content signal|.

Both predicted the corresponding bootstrapped difference to be positive.

5.6 Data quality

560/560 calls succeeded: **0 call failures, 0 parse failures, 100% strict JSON**. The served model matched the pinned ID on every call. Eight integrity checks — duplicate IDs, call and parse failures, served-model match, exact counterbalance, full repetitions, independently re-derived semantic mapping, and displayed-text consistency — all passed **before** any statistic was computed.

6. Study 2 results

6.1 Both hypotheses are rejected, in the opposite direction

Model	mean position effect	mean content signal	sanity-control accuracy
GPT-OSS 20B	0.442 [0.292, 0.600]	0.275 [0.158, 0.408]	0.975
GPT-OSS 120B	0.925 [0.858, 0.983]	0.058 [0.000, 0.125]	1.000
<i>chance (coin-flip, same design)</i>	<i>0.176 (p95 0.242)</i>	<i>0.176 (p95 0.242)</i>	—

Paired bootstrap over the 12 shared items:

- **H1 rejected.** $\Delta|\text{position}|$ (20B – 120B) = **−0.483**, 95% CI [−0.650, −0.317]. Predicted positive; observed strongly negative.
- **H2 rejected.** $\Delta|\text{content}|$ (120B – 20B) = **−0.217**, 95% CI [−0.325, −0.117]. Predicted positive; observed strongly negative.

The larger model was **more** position-dominated and carried **less** order-invariant content signal. Both intervals exclude zero, on the wrong side of the prediction.

gpt-oss-120b returned `p_first = 1.0` on **11 of 12** balanced items (0.9 on the twelfth) and `p_second = 0.0` on **8 of 12**; at the trial level it chose the first-displayed option on **231 of 240 trials (96.25%)**. The mean |position effect| is 0.925 rather than 1.0 because four items (p03, p10, p11, p12) retained a small residual content signal. Figure C shows this is not driven by a few items — every one of its twelve items sits at or above 0.7, far outside the chance band.

Note also that GPT-OSS 20B's content signal (0.275) is only modestly above the chance level (0.176). Neither arm shows a strong order-invariant preference structure; they differ mainly in how completely position fills the gap.

6.2 The dissociation: it is not a comprehension failure

On the sanity controls — where one option is degenerate — gpt-oss-120b scored **100% accuracy with a position effect of exactly 0.000**, and 20B scored 97.5%.

So the larger model ignores display order **completely** when one option is plainly invalid, and follows it **almost completely** when both are reasonable. The controls therefore exclude a *global* prompt-reading or option-parsing failure: the model can override position when the semantic distinction is sufficiently decisive (Figure D). They do **not** establish that it discriminates the subtler content differences among the balanced task pairs — only that its position-following there is not a blanket inability to process the prompt.

What it does **not** show is that no preference exists. A model may have dispositions our protocol cannot surface.

6.3 Exploratory: weak content, strong position

Pre-specified as exploratory, not confirmatory (D-27). Spearman correlation between |content signal| and |position effect| across items:

rho	p	n
0.664	0.0002	24

GPT-oss-120B	0.0004	12
GPT-oss-20B	0.79	12

Pooled, items with weaker content association show stronger position effects. But the association is carried almost entirely by the 120B arm and is absent in the 20B arm, and the pooled figure mixes two models with very different distributions. We report it as a hypothesis for future work, not a finding.

6.4 Study 2b — provider-pinned replication

The control Study 2 claimed was incomplete. OpenRouter's `allow_fallbacks: false` pins the served *model*, not the upstream inference provider. OpenRouter lists 12 upstream endpoints for `gpt-oss-20b` and 20 for `gpt-oss-120b`, and the original Study 2 run neither pinned nor recorded which one served each call. The serving stack was therefore **not** held fixed, contrary to what Study 2's write-up implied (D-33).

We repaired this rather than only caveating it. Study 2b re-runs the identical design with both arms pinned to a single upstream provider (**Groq**, which serves both models) via `provider.only`, recording the served provider on every record. All 560 calls report `served_provider = Groq`; the pin is asserted before any statistic is computed. The original dataset is preserved unchanged.

	20B unpinned	20B Groq	120B unpinned	120B Groq
mean position effect	0.442	0.608	0.925	0.942
mean content signal	0.275	0.208	0.058	0.058
control accuracy	0.975	1.000	1.000	1.000

hypothesis	unpinned Δ	Groq-pinned Δ	verdict
H1 position (20B – 120B)	–0.483 [–0.650, –0.317]	–0.333 [–0.492, –0.192]	rejected in both
H2 content (120B – 20B)	–0.217 [–0.325, –0.117]	–0.150 [–0.217, –0.092]	rejected in both

The finding replicates under provider control. Both hypotheses remain rejected in the same direction, with intervals excluding zero.

Two things are worth noting. First, the effect is *smaller* under pinning (Δ |position| $-0.483 \rightarrow -0.333$), mostly because 20B's position effect rose from 0.442 to 0.608 on Groq — so upstream provider identity does measurably affect these numbers, which is precisely why the original uncontrolled run should not have been described as provider-controlled. Second, the exploratory content-vs-position association strengthens under pinning (pooled $p = -0.869$, and -0.632 for 20B where it was null before), but it remains exploratory.

Study 2b is the version we treat as the controlled result. Study 2 is retained as the original, uncontrolled run.

6.5 Study 3 — does our own instruction create the artefact?

Both studies so far tell the model "There is no correct answer and no choice is more helpful than any other." That sentence is meant to avoid pushing toward a "right" answer, but it may also assert that the alternatives are *equivalent* — and a model that perceives a tie has to break it somehow. The sceptical reading is that **our elicitation instruction manufactured the artefact we measured**.

Study 3 tests exactly that. Everything from Study 2b is held fixed — same models, same Groq upstream pin, same 12 items and 2 controls, same exact counterbalancing, same sampling — and one sentence is deleted. A programmatic

diff of the two configs confirms only `study_id` and `system_prompt` differ. RQ4, H3 and H4 were written into `DECISIONS.md` (D-38) before the run. 560/560 calls, zero failures.

model	metric	indifference cue	cue removed	paired diff	95% CI
120B	position	0.942	0.925	+0.017	[−0.025, +0.067]
120B	content	0.058	0.075	+0.017	[−0.025, +0.067]
20B	position	0.608	0.483	+0.125	[+0.025, +0.225]
20B	content	0.208	0.317	+0.108	[−0.017, +0.233]

The explicit-indifference-cue explanation is not supported for the 120B model. Extreme positional responding persisted without the cue: |position| went 0.942 → 0.925, a paired change of +0.017 [−0.025, +0.067]. We did not pre-specify an equivalence margin, so this is not positive evidence of exact equivalence; the defensible statement is that **whatever effect deleting that sentence had, the 120B model remained extremely position-dominated**.

Note also what was *not* manipulated. We removed one specific sentence, not every feature of the protocol that might encourage default or tie-breaking behaviour — the forced binary choice, the JSON schema, and the welfare-neutral item set all remain.

For the 20B model the change was in the pre-specified direction, |position| 0.608 → 0.483, and its unadjusted 95% CI excluded zero (+0.125 [+0.025, +0.225]), providing **suggestive** evidence that the cue increased positional susceptibility. Its content signal moved as H4 predicted (0.208 → 0.317) but that interval includes zero.

The two models may differ. The model × framing interaction on |position| is −0.108 [−0.208, −0.008]. This is **similarly suggestive**: it points to framing mattering for the smaller model and not detectably for the larger one, i.e. elicitation framing may interact with model identity rather than exerting a uniform effect.

Multiplicity. Four hypothesis tests plus the interaction. The 20B effect and the interaction only just exclude zero and **would not survive a strict correction across the five reported tests**; both are reported as suggestive throughout. The 120B result does not depend on a threshold, since it is a small estimate with an interval spanning zero rather than a marginal exclusion.

Why this matters for the track. The finding is not "LLMs have position bias", which is established. It is that **an ostensibly innocuous elicitation instruction measurably modulated positional responding in one model and not the other** — so preference-measurement design is itself a variable that can create or amplify the artefact, and a protocol's framing needs reporting as carefully as its sampling parameters.

6.6 Relationship to Study 1

These are **separate experiments** and their numbers are not pooled. Study 1 randomised order probabilistically across four methods; Study 2 counterbalanced exactly within one method. The sample structures differ and the metrics are computed differently.

Descriptively, however, the phenomenon is the same one, and it now spans five models. GPT-OSS figures below are from **Study 2b (provider-pinned)**, the controlled run:

model	study	protocol	position measure
GPT-OSS 120B	2b (pinned)	exact counterbalance	position effect 0.942
Qwen-2.5-7B	1	randomised order	position effect +0.76 pooled; pairwise degenerate
GPT-OSS 20B	2b (pinned)	exact counterbalance	position effect 0.608
Llama-3.1-8B	1	randomised order	+0.51 pooled
Gemini-3.1-flash-lite	1	randomised order	+0.29 pooled

The Study 1 and Study 2 columns are **not directly comparable** — different protocols, different estimators, different item coverage — and the table is descriptive only. What it shows is that substantial position dependence appeared in every model tested, across two providers and four model families.

7. Discussion

Cross-method agreement alone is not sufficient. Against a matched permutation null, Study 1 found convergence clearly above null for Gemini, **ambiguous** for Llama (nominally significant on its fuller 4-method basis, $p = 0.014 / 0.033$, but not on the matched basis and not after Bonferroni correction), and not detectable for Qwen. The tempting reading — "the methods disagree" — is wrong for at least one model. The decomposition shows Qwen's pairwise measure supplied no detectable order-invariant signal against which cross-method agreement could be interpreted. A near-chance convergence result looks identical whether the methods genuinely conflict or one of them carries no signal, and convergence statistics alone cannot tell those apart.

The converse warning is just as important. Several methods can agree because they share a nuisance variable rather than because they track a common signal. All four of our methods present two labelled options to the same model through text; a position bias affects all of them, and would produce agreement that looks like convergent validity.

Position diagnostics are cheap and informative. Reporting $P(X \mid X \text{ first})$ against $P(X \mid X \text{ second})$ costs nothing beyond counterbalancing the order, and it is what reveals whether an aggregate estimate is being driven by display order. Without it, a standard pairwise elicitation produced for Qwen a confident-looking twelve-item preference vector that reproduced the random display-order draw *exactly*, on every item.

Sanity checks do not catch this. Both Qwen (Study 1) and GPT-OSS 120B (Study 2) rejected degenerate options correctly — 120B with 100% accuracy and a position effect of exactly zero — while being almost entirely position-driven on balanced items. A model can look completely healthy on validity controls and still yield preference measurements that are pure artefact. Control items test comprehension; they do not test whether a preference was measured.

Order imbalance can obscure agreement, not only manufacture it. Equal- weighting the two display orders raised Llama's self-report/pairwise correlation from $\rho = -0.007$ to $+0.524$ (§4.3), consistent with position contributing to their apparent disagreement. For Qwen the adjusted measure is a constant zero, so a correlation does not exist rather than being low. This is exploratory, covers one method pair, and does not extend to the three-method figures in §4.1 — but it shows the familiar worry has a mirror image: leaving position in an estimate can *understate* agreement as well as fabricate signal.

Elicitation framing may interact with model identity. Deleting one sentence — "there is no correct answer and no choice is more helpful than any other" — left the 120B model extremely position-dominated ($0.942 \rightarrow 0.925$, $+0.017$ with an interval spanning zero), so the explicit-indifference-cue explanation is not supported there. The 20B model moved in the predicted direction ($0.608 \rightarrow 0.483$, CI excluding zero), and the model $\times$ framing interaction was -0.108 [-0.208 , -0.008]. Both of the latter are **suggestive** rather than established: neither survives strict correction across the five reported tests. The reasonable reading is that framing may interact with model identity rather than exerting a uniform effect, which makes prompt framing a variable worth reporting as carefully as temperature.

The capability explanation did not survive testing. After Study 1 the natural hypothesis was that position dominance reflects a small model's inability to compare options, and we recorded it as a pre-specified hypothesis before collecting any Study 2 data (D-27). Study 2b rejected it in the opposite direction. With family, prompts, sampling and the upstream inference provider all held fixed, the **larger** model was substantially more position-dominated ($\Delta|\text{position}| = -0.333$ [-0.492 , -0.192]) and carried **less** order-invariant content signal ($\Delta|\text{content}| = -0.150$ [-0.217 , -0.092]). The original, provider-uncontrolled Study 2 gave larger estimates (-0.483 and -0.217); we quote the controlled ones.

We are careful about what this licenses. Scale remains confounded with training compute, data mixture and post-training even within a family, so this is not evidence that scale *causes* position dominance. The defensible claim is narrow and negative: a controlled within-family, within-provider comparison found positional susceptibility **higher** in the larger model, which is inconsistent with the capability account as we stated it. Whatever governs positional susceptibility, it is not simply "bigger models compare better."

Large model-to-model differences were observed. Across five models, measurement behaviour varied substantially between models. We have not run a formal variance decomposition, so we do not claim model identity is *the* dominant factor — only that the between-model spread was large. Any claim of the form "LLMs prefer X" that rests on one model and one procedure is, on this evidence, unsafe.

Self-report was the weakest method. On both open-weight models in Study 1, direct self-report agreed least with the behavioural methods, and on Llama it had the lowest content signal (0.079) alongside the highest position effect (+0.841). This is also why we never asked any model to explain its own position bias: using the least reliable method in the study to explain the study's central finding would be self-undermining (D-30).

On Study 1's contradicted secondary hypothesis. Stronger pairwise preferences were *less*, not more, method-stable. One candidate explanation is that extreme pairwise scores arise partly from saturation ($P(A)$ at 0 or 1), which position lock-in produces as readily as strong preference. A hypothesis generated by the data, not a result.

No ontological conclusion. Nothing here bears on consciousness, sentience, welfare, phenomenology or genuine internal preference. All three studies measure whether measurement procedures agree with each other and whether their output survives a nuisance-variable control. Consistency does not establish that anything is preferred; inconsistency does not establish that nothing is.

8. Conclusion

We measured apparent preferences over 12 welfare-neutral task pairs with four elicitation procedures across three model families, then ran a controlled within-family scale comparison on two more, and finally tested whether our own prompt framing drives the effect.

Against a matched permutation null, cross-method convergence exceeded chance in one model (Gemini, $p \leq 0.002$), was suggestive but not established in a second (Llama), and was not detectable in the third (Qwen). The model differences coincided with large differences in order-invariant signal. In one model the standard pairwise procedure recovered nothing but display order, while still passing validity controls; in another, equal-weighting the display orders raised an apparently null cross-method correlation to +0.524, so order imbalance can obscure agreement as well as fabricate it.

A follow-up with pre-specified hypotheses (Study 2b), holding family and upstream provider fixed, found the larger model *more* position-dominated, rejecting the capability explanation we had proposed. Removing the explicit indifference cue left GPT-OSS 120B extremely position-dominated, while the 20B model showed a suggestive reduction in positional susceptibility.

The actionable recommendation is small and cheap: **counterbalance option order and report position-conditioned choice rates alongside aggregate preference frequencies**. Without position-conditioned rates, an aggregate estimate cannot rule out a display-order artefact. These diagnostics detect order sensitivity; they do not establish that a genuine preference was measured.

Appendix — Limitations, Dual-Use and Ethical Considerations

See [limitations.md](#), which is part of this report. Key points: convergence cannot distinguish a shared training-induced bias from a common signal (every method queries the same model through text); disagreement does not establish absence of preference; there is no ground truth; persona and post-training effects plausibly shape every number here; Method D has no Gemini data and Methods C/D were not framing-tested; Study 2 covers one method only, so it does not show that the *other* three methods behave the same way under exact counterbalancing; scale is confounded with training differences even within a family; results are specific to temperature 1.0, English, and one system prompt; and 10–12 items keeps every interval wide.

Figures

Study 1

- **Fig 1** — item × method normalised score heatmap, per model.

- **Fig 2** — method-pair Spearman correlation matrix, per model.
- **Fig 3** — [pairwise strength] vs cross-method disagreement. Marked INVALID for Qwen, whose pairwise measure is degenerate.
- **Fig 4** — framing sensitivity (Llama, Qwen).
- **Fig 5** — cross-method convergence vs mean [content effect], matched subset.

Study 2 (results/followup/figures/)

- **Fig A** — position vs content plane, one point per item, both models.
- **Fig B** — model-level comparison with bootstrap CIs and the chance level.
- **Fig C** — per-item position effect, showing the aggregate is not driven by a few items.
- **Fig D** — position dominance on balanced items versus sanity controls.

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All four references were retrieved and their identifiers verified during this sprint. No reference in this report is reconstructed from memory alone.